Classical ideal gas: microcanonical approach.

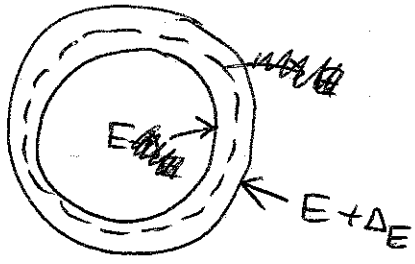
$$H = \sum_{j=1}^{N} \left[\frac{\vec{p}_j^{\,2}}{2m} + U(\vec{q}_j) \right]$$

↳ walls of container

Microcanonical ensemble: all fixed $\begin{cases} \text{number of particles } N \\ \text{volume } V \\ \text{energy } E \end{cases}$

(1) # of allowed microstates $\Omega(E,N,V)$

<u>Important</u>: the states must be counted with some interval Δ_E



let $\tilde{\Omega}(E,N,V)$ be the number of states with total energy $\leq E$.

Then $\Omega(E,N,V) = \frac{d\tilde{\Omega}}{dE} \cdot \Delta_E$

$$E = \sum_j \frac{\vec{p}_j^{\,2}}{2m} \Rightarrow \sum_j \vec{p}_j^{\,2} = 2mE \Rightarrow$$ 3N-dimensional sphere of radius $R_E = \sqrt{2mE}$

$$\tilde{\Omega}(E,N,V) = V^N \underbrace{V_E^{(3N)}}_{\text{volume of 3N-dimensional sphere}}$$

$$V_E^{(3N)} = C_{3N} \cdot R_E^{3N}$$

numerical coefficient

Thus $\Omega(E,N,V) = \frac{d}{dE}\left(V^N V_E^{(3N)}\right) \cdot \Delta_E =$

$$= \frac{d}{dR_E}\left(V^N V_E^{(3N)}\right) \underbrace{\frac{dR_E}{dE}}_{\sqrt{\frac{m}{\ }}} \Delta_E$$

$$\Omega(E,N,V) = V^N C_{3N}\, 3N\, R_E^{3N-1} \sqrt{\frac{m}{2E}}\, \Delta_E$$

Note that $\Omega(E,N,V)$ is dimensionful:

$$\Omega(E,N,V) = 3N\, C_{3N}\, V^N R_E^{3N} \cdot \left(\frac{\Delta_E}{2E}\right)$$

$$\dim[\Omega] = \left(\dim[V]\right)^N \cdot \left(\dim[R_E]\right)^{3N} = (\text{Length})^{3N} (\text{Momentum})^{3N}$$

$\Rightarrow$ to get dimensionless number of states we have to divide Ω by something with dimension $(\text{Length} \cdot \text{Momentum})^{3N}$.

It turns out that that is provided by Planck's constant $h = 2\pi\hbar$: $\dim[h] = \text{Length} \cdot \text{Momentum}$.

$\Rightarrow$ finally $\Omega(E,N,V) = 3N\, C_{3N} \left(\frac{V^{1/3} R_E}{2\pi\hbar}\right)^{3N} \left(\frac{\Delta_E}{2E}\right)$

Entropy $\sigma(E,N,V) = \ln \Omega(E,N,V) =$

$$= \ln(3N) + \ln(C_{3N}) + 3N \ln\left(\frac{V^{1/3} R_E}{2\pi\hbar}\right) + \ln\left(\frac{\Delta_E}{2E}\right)$$

$(V = L^3)$

$$= N\left\{ 3 \ln \frac{L R_E}{2\pi\hbar} + \frac{1}{N} \ln(C_{3N}) + \frac{1}{N} \ln(3N) + \frac{1}{N} \ln\left(\frac{\Delta_E}{2E}\right)\right\}$$

(this will turn to be important: $C_{3N} \sim (3N!)$)

ϕ for $N \to \infty$ neglect!!!

$$\frac{1}{T} = \left(\frac{\partial \sigma}{\partial E}\right)_{N,V}$$

$$\frac{1}{\tau} = \left(\frac{\partial \sigma}{\partial R_E}\right) \frac{\partial R_E}{\partial E} = \sqrt{\frac{m}{2E}} \; \frac{3N}{R_E} = 3N\sqrt{\frac{m}{2E}} \; \frac{1}{\sqrt{2mE}} =$$

$$= \frac{3N}{2E}$$

$$\Rightarrow \quad E = \frac{3N}{2}\tau \quad \leftarrow \text{ this is ideal gas law!}$$

$$\left(E = \frac{3N}{2} k_B T \text{ in standard units} \right)$$

$\frac{1}{2}\tau = \frac{1}{2} k_B T$ per degree of freedom!

(2) What is probability $P(\vec{p}_1)$ to find a particle of momentum $\vec{p}_1$ in the gas?

Total energy of N particles $= E$
Energy of one particle $= \frac{\vec{p}_1^{\,2}}{2m}$

Energy of $N-1$ particles is $E - \frac{\vec{p}_1^{\,2}}{2m}$.

That particle can be anywhere inside V

$$\Rightarrow \quad P(\vec{p}_1) = \frac{\Omega\left(E - \frac{\vec{p}_1^{\,2}}{2m}, N-1, V\right)}{\Omega(E, N, V)} \cdot V =$$

$$= V \cdot \frac{3(N-1)\, C_{3(N-1)}\, V^{N-1}\, R_{E-p^2/2m}^{3(N-1)} \; \frac{\Delta E}{2\left(E - \frac{p^2}{2m}\right)}}{3N\, C_{3N}\, V^{N}\, R_E^{3N} \; \frac{\Delta E}{2E}} =$$

But $$\frac{R_E^{3N}}{E} = \frac{\left(\sqrt{2mE}\right)^{3N}}{E} = (2m)^{\frac{3N}{2}} E^{\frac{3N}{2}-1}$$

$$\frac{R^{3(N-1)}_{E-p^2/2m}}{E-\frac{p^2}{2m}} = \frac{\left(\sqrt{2m\left(E-\frac{p^2}{2m}\right)}\right)^{3(N-1)}}{E-\frac{p^2}{2m}} = (2m)^{\frac{3}{2}(N-1)}\left(E-\frac{p^2}{2m}\right)^{\frac{3}{2}(N-1)}$$

$$= (2m)^{\frac{3}{2}(N-1)}\, E^{\frac{3N-5}{2}}\left(1-\frac{p^2}{2mE}\right)^{\frac{3N-5}{2}}$$

$$P(\vec{p}) = \frac{C_{3(N-1)}}{C_{3N}} \frac{(2m)^{\frac{3}{2}(N-1)}\, E^{\frac{3N-5}{2}}\left(1-\frac{p^2}{2mE}\right)^{\frac{3N-5}{2}}}{(2m)^{\frac{3}{2}N}\, E^{\frac{3N-2}{2}}} =$$

$$= \frac{C_{3(N-1)}}{C_{3N}} \frac{1}{(2mE)^{3/2}}\left(1-\frac{p^2}{2mE}\right)^{\frac{3N-5}{2}}$$

* But for $N\to\infty$: $\left(1-\frac{p^2}{2mE}\right)^{\frac{3N}{2}-\frac{5}{2}} \cong \left(1-\frac{p^2}{2mE}\right)^{\frac{3N}{2}} =$

$$\cong \exp\left(-\frac{3N}{2}\frac{p^2}{2mE}\right) \quad !$$

Using $C_{3N} = \dfrac{\pi^{\frac{3N}{2}}}{\left(\frac{3N}{2}\right)!} \Rightarrow C_{3(N-1)} = \dfrac{\pi^{\frac{3}{2}(N-1)}}{\left(\frac{3}{2}(N-1)\right)!}$

$$\Rightarrow \frac{C_{3(N-1)}}{C_{3N}} = \pi^{-3/2}\frac{\left(\frac{3}{2}N\right)!}{\left(\frac{3}{2}(N-1)\right)!}$$

* Note:

$$\left(1-\frac{p^2}{2mE}\right)^{\frac{3N}{2}} = \left(1-\frac{p^2}{2mT(3N/2)}\right)^{\frac{3N}{2}} \xrightarrow[N\to\infty]{} \exp\left(-\frac{p^2}{2mT}\right).$$

Use Stirling's formula:

Consider $\ln \frac{C_{3(N-1)}}{C_{3N}} = \ln \pi^{-3/2} + \ln\left(\frac{3N}{2}\right)! -$

$$- \ln\left(\frac{3(N-1)}{2}\right)! = \ln \pi^{-3/2} + \frac{3N}{2} \ln \frac{3N}{2} - \frac{3N}{2}$$

$$- \left[\frac{3N-3}{2} \ln \frac{3N-3}{2} - \frac{3N-3}{2} \right] = \ln \pi^{-3/2} + \frac{3N-3-(3N)}{2}$$

$$+ \frac{3N}{2} \ln \frac{3N}{2} - \frac{3N}{2} \underbrace{\left(1 - \frac{3}{3N}\right)}_{\left(1 - \frac{1}{N}\right)} \ln \frac{3N-3}{2} =$$

$$= \ln \pi^{-3/2} - \frac{3}{2} + \frac{3N}{2} \ln \frac{3N}{3N-3} - \frac{3N}{2}\left(-\frac{1}{N}\right) \ln \frac{3N-3}{2}$$

$$+ \frac{3}{2} \ln\left(\frac{3N-3}{2}\right) =$$

$$= \ln \pi^{-3/2} - \frac{3}{2} + \frac{3N}{2} \ln\left(1 + \frac{1}{N}\right) + \frac{3}{2} \ln \frac{3N}{2}$$

$$= \ln \pi^{-3/2} - \frac{3}{2} + \frac{3N}{2} \frac{1}{N} + \ln\left(\frac{3N}{2}\right)^{3/2}$$

$$= \ln \left(\frac{3N}{2\pi}\right)^{3/2} + o\left(\frac{1}{N}\right).$$

$$\Rightarrow \quad \frac{C_{3(N-1)}}{C_{3N}} = \left(\frac{3N}{2\pi}\right)^{3/2}$$

$$\Rightarrow \quad P(\vec{p}) = \left(\frac{3N}{4\pi m E}\right)^{3/2} \exp\left(-\frac{3N}{2} \frac{p^2}{2mE}\right)$$

But $E = \frac{3N}{2}\tau \Rightarrow \frac{3N}{2E} = \frac{1}{\tau} \Rightarrow$

$$P(\vec{p}) = \frac{1}{(2\pi m \tau)^{3/2}} \exp\left(-\frac{\vec{p}^{\,2}}{2m\tau}\right)$$

(Normalized) Maxwell-Boltzmann distribution!!!

So generally the number of states is given by

$$\Omega(E,N,V) = \int_{\mathcal{H}=E} \frac{d\vec{q}_1\, d\vec{p}_1}{(2\pi\hbar)^3} \frac{d\vec{q}_2\, d\vec{p}_2}{(2\pi\hbar)^3} \cdots \frac{d\vec{q}_N\, d\vec{p}_N}{(2\pi\hbar)^3}$$

$$= \int_{\mathcal{H}=E} \prod_{j=1}^{N} \frac{d\vec{p}_j\, d\vec{q}_j}{(2\pi\hbar)^3}$$

The integral over constant-energy surface in 6N-dimensional phase space of the system.

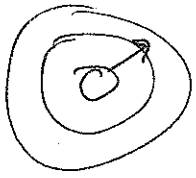
(For large $N \gg 1$ the difference between surface and volume is negligible: that difference goes as $\frac{1}{N} \to 0$.)

do it

Volume of * hypersphere

n-dimensional

$$\sum_{i=1}^{n} x_i^2 = r^2 \qquad V_n = c_n r^n \qquad \left(c_2 = \pi,\ c_3 = \frac{4\pi}{3}\right)$$



$$V_n = \int_0^r S_n(r)\,dr \Rightarrow S_n = \frac{d}{dr} V_n = n c_n r^{n-1}$$

We want to find $\underline{c_n}$:

$$J = \int_0^\infty e^{-ar^2} S_n(r)\,dr = n c_n \int_0^\infty e^{-ar^2} r^{n-1}\,dr =$$

$$= (r^2 = z) \quad = \frac{n c_n}{2} \int_0^\infty e^{-az} z^{\frac{n}{2}-1} z^{-1/2}\,dz = \frac{n c_n}{2} \int_0^\infty e^{-az} z^{\frac{n}{2}-1}\,dz$$

$r = z^{1/2}$
$dr = \frac{1}{2} z^{-1/2} dz$

$$= \frac{1}{2} n c_n \int_0^\infty e^{-t} t^{\frac{n}{2}-1}\,dt\ a^{-\left(\frac{n}{2}-1\right)-1}$$

$z = t/a$

$$= \frac{n c_n}{2a^{n/2}} \underbrace{\int_0^\infty e^{-t} t^{\frac{n}{2}-1}\,dt}_{\Gamma\left(\frac{n}{2}\right)}$$

because $\int_0^\infty e^{-t} t^b\,dt = \Gamma(b+1)$

On the other hand $J = \int_0^\infty e^{-ar^2}\,dV_n =$

$$= \int_{-\infty}^{\infty} e^{-a(x_1^2 + x_2^2 + \dots + x_n^2)}\,dx_1\,dx_2 \dots dx_n =$$

$$= \left(\int_{-\infty}^{\infty} e^{-a x_1^2} dx_1\right)\left(\int e^{-a x_2^2} dx_2\right) \dots \left(\int e^{-a x_n^2} dx_n\right) =$$

$$= \left(\int e^{-ax^2} dx\right)^n = \left(\sqrt{\frac{\pi}{a}}\right)^n = \left(\frac{\pi}{a}\right)^{n/2}$$

$$\Rightarrow \quad 1 = \left(\frac{\pi}{a}\right)^{n/2} = \frac{n C_n}{2 a^{n/2}} \Gamma\left(\frac{n}{2}\right)$$

$$\Rightarrow \quad C_n = \frac{2 \pi^{n/2}}{n \Gamma\left(\frac{n}{2}\right)} = \frac{\pi^{n/2}}{\frac{n}{2} \Gamma\left(\frac{n}{2}\right)} = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} \quad !$$

$n=2$ $\quad C_n = \dfrac{\pi}{\Gamma(2)} = \pi$

$n=3$ $\quad C_n = \dfrac{\pi^{3/2}}{\Gamma\left(\frac{3}{2}+1\right)} = \dfrac{\pi^{3/2}}{\frac{3}{2}\Gamma\left(\frac{3}{2}\right)} = \dfrac{\pi^{3/2}}{\frac{3}{2}\cdot\Gamma\left(\frac{1}{2}+1\right)} = \dfrac{\pi^{3/2}}{\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)} = \dfrac{4\pi^{3/2}}{3\sqrt{\pi}} = \dfrac{4\pi}{3}$

$$\Rightarrow \text{for } n = 3N \qquad C_n = C_{3N} = \frac{\pi^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)}$$